MIL-C-25 6C
21 June 1968
SUPERSEDING
MIL-C-25 6B
8 February 1960

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PAPER-DIELECTRIC, DIRECT CURRENT (HERMETICALLY SEALED IN METAL CASES), STYLES CP67 and CP69

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-25.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

REQUIREMENTS:

Dimensions: See figure 1 and tables I and II.

Case: Magnetic or nonmagnetic material, uninsulated.

Design and construction: Capacitors shall be of the design, construction and physical dimensions as specified in figure 1 and tables I and II.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC voltage rating: See table I.

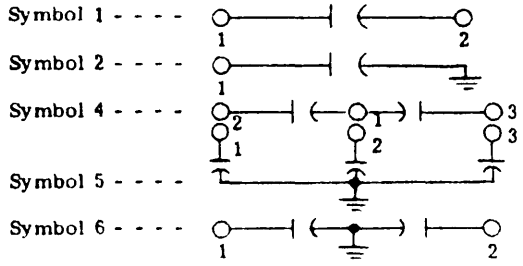
Operating temperature range:
-55° to +85° C.

Vibration grade: 1 (10 to 55 Hz, incl.)

Terminal:

Symbol B - - - - Solder lug (nonremovable).

Circuit diagram:



Voltage derating with temperature: See figure 2.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
Terminal to terminal - As specified in table IV.
Terminal to case (when case is not a terminal) - As specified in table IV. Capacitors shall withstand specified voltages with no permanent damage or open- or short-circuiting.

Insulation resistance: Method 302 of MIL-STD-202.
At 25° C:
Terminal to terminal - As specified in table V.
Terminal to case (when case is not a terminal) - Shall exceed 3,000 megohms.
At 85° C (characteristics E and F):
Terminal to terminal - As specified in table V.

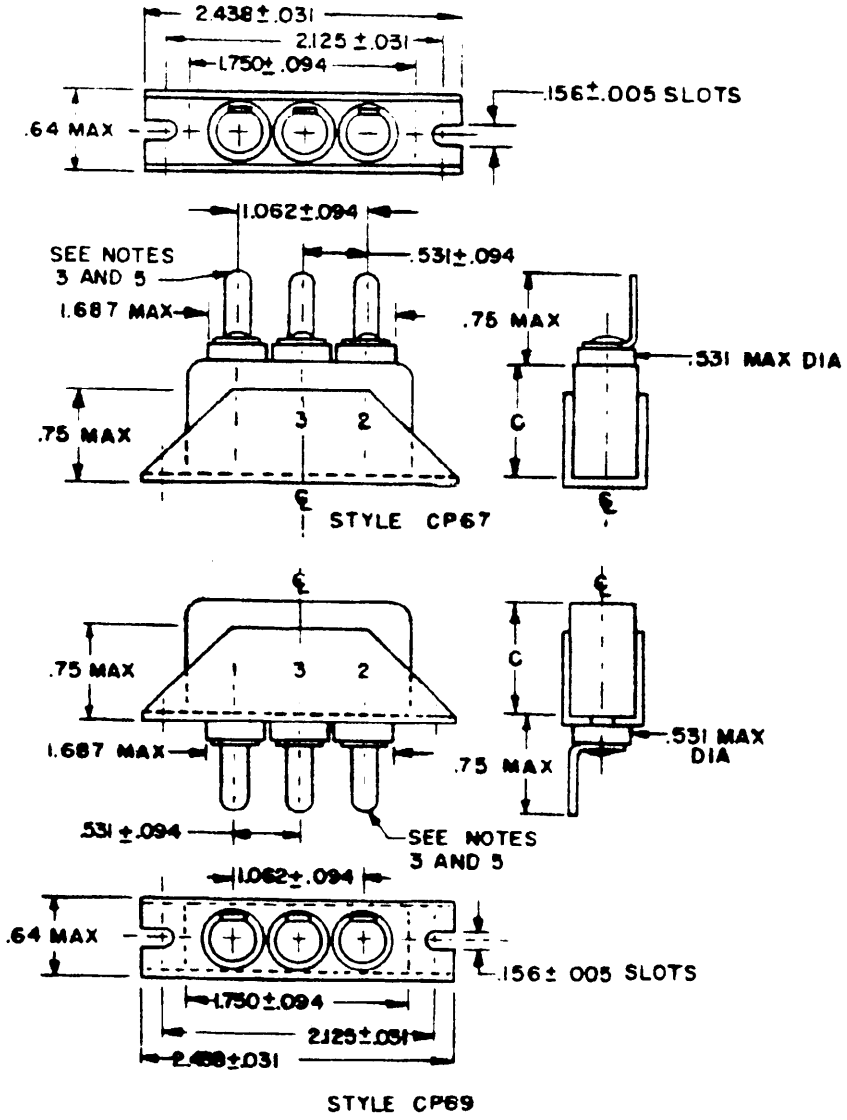
Capacitance: Method 305 of MIL-STD-202. Measured capacitance shall be within the tolerance specified in table I.

Dissipation factor: When measured as specified in MIL-C-25, shall not exceed 1 percent.

Barometric pressure (flashover):
Method 105 of MIL-STD-202.
Test condition - B (50,000 feet).
Test potential - 125 percent of rated dc voltage, applied for a minimum of 1 minute, without visible damage, external flashover, or open- or short-circuiting.

FSC 5910

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TABLE I. Styles CP67 and CP69 characteristics and dimensions.

Type designation 1/	Dc voltage rating (see fig. 2)	Capacitance	Capacitance tolerance 2/	Characteristics	Vibration grade	Case size	
						Characteristic E	Characteristic F
	Volts	μf					
CP6-B-EF503K1	600	0.05	K	E	1	B1	...
CP6-B-EF104K1	600	0.1	K	E	1	B1	...
CP6-B-EF254K1	600	0.25	K	E	1	B2	...
CP6-B-EF504K1	600	0.5	K	E	1	B3	...
CP6-B--F105K1	600	1	K	E, F	1	B5	B4
CP6-B-EG103K1	1,000	0.01	K	E	1	B1	...
CP6-B-EG203K1	1,000	0.02	K	E	1	B1	...
CP6-B-EG503K1	1,000	0.05	K	E	1	B1	...
CP6-B--G104K1	1,000	0.1	K	E, F	1	B2	B1
CP6-B-EG254K1	1,000	0.25	K	E	1	B3	...
CP6-B--G504K1	1,000	0.5	K	E, F	1	B5	B4
CP6-B-EF503V1	600	0.05-0.05 3/	V	E	1	B1	...
CP6-B-EF104V1	600	0.1 -0.1	V	E	1	B2	...
CP6-B-EF254V1	600	0.25-0.25	V	E	1	B3	...
CP6-B--F504K1	600	0.5 -0.5	V	E, F	1	B5	B4
CP6-B-EG103V1	1,000	0.01-0.01	V	E	1	B1	...
CP6-B-EG203V1	1,000	0.02-0.02	V	E	1	B1	...
CP6-B--G503V1	1,000	0.05-0.05	V	E, F	1	B2	B1
CP6-B--G104V1	1,000	0.1 -0.1	V	E, F	1	B3	B2
CP6-B--G254V1	1,000	0.25-0.25	V	E, F	1	B5	B4
CP6-B5-F503V1	600	0.05-0.05-0.05	V	E, F	1	B2	B1
CP6-B5EFP04V1	600	0.1-0.1-0.1	V	E	1	B2	...
CP6-B5-F254V1	600	0.25-0.25-0.25	V	E, F	1	B5	B4
CP6-B5EG103V1	1,000	0.01-0.01-0.01	V	E	1	B1	...
CP6-B5EG203V1	1,000	0.02-0.02-0.02	V	E	1	B1	...
CP6-B5EG503V1	1,000	0.05-0.05-0.05	V	E	1	B2	...
CP6-B5EG104V1	1,000	0.1 -0.1 -0.1	V	E	1	B4	...

1/ Complete type designation shall include additional symbols to complete style designation, indicate circuit (where applicable) and characteristics (where applicable).

2/ Capacitance tolerance in percent: K, ±10; V, +20, -10.

3/ For multiple-unit capacitors, the watt-second rating is the sum of the watt-second ratings of the component sections.

TABLE II. Case dimensions.

Case size	Dimensions C±.062
B1 - - - - - -	1.062
B2 - - - - - -	1.500
B3 - - - - - -	2.000
B4 - - - - - -	2.500
B5 - - - - - -	2.750

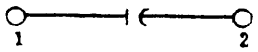
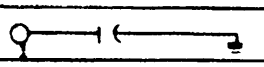
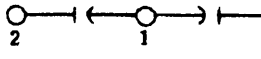
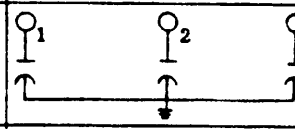
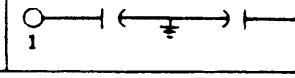
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tication sheets shall be as specified on applicable specifi

TABLE III. Millimeter equivalent of decimal inches.

In.	mm	In.	mm	In.	mm
.005	.13	.640	16.26	1.750	44.45
.031	.79	.750	19.05	2.000	50.80
.094	2.39	1.062	26.99	2.125	53.98
.156	3.96	1.500	38.10	2.438	61.93
.531	13.49	1.687	42.85	2.750	69.85

TABLE IV. Dielectric withstanding voltage test.

Circuit symbol	Circuit diagram	Test	Test connections	Test voltage (percent of rated dc voltage)	Time test applied (minutes)
1		Terminal to terminal	1 to 2	200 [1/]	1 [2/]
		Terminal to case	1 and 2 to case	400 [5/]	1 [3/4/]
				200 + 1,000 [6/] volts	1 [3/4/]
2		Terminal to terminal	1 to case	200 [1/]	1 [2/]
4		Terminal to terminal	1 to 2 and 3 together	200 [1/]	1 [7/]
			2 to 3	200	1 [7/]
		Terminal to case	1, 2, and 3 to case	400	1 [3/4/]
				200 + 1,000 volts	1 [3/4/]
5		Terminal to terminal	1, 2, and 3 to case	200	1 [7/]
			1 and 2 to 3	200	1 [7/]
			1 to 2	200	1 [7/]
6		Terminal to terminal	1 and 2 to case	200	1 [7/]
			1 to 2	200	1 [7/]

[1/] 175 percent of rated dc voltage after temperature-and immersion-cycling and moisture resistance tests.

[2/] For quality conformance inspection and at the option of the supplier, either 200 percent of rated dc voltage shall be applied for 15 seconds or 250 percent of rated dc voltage for not less than 1 second.

[3/] For quality conformance inspection, the period of test-voltage application shall be 1 second.

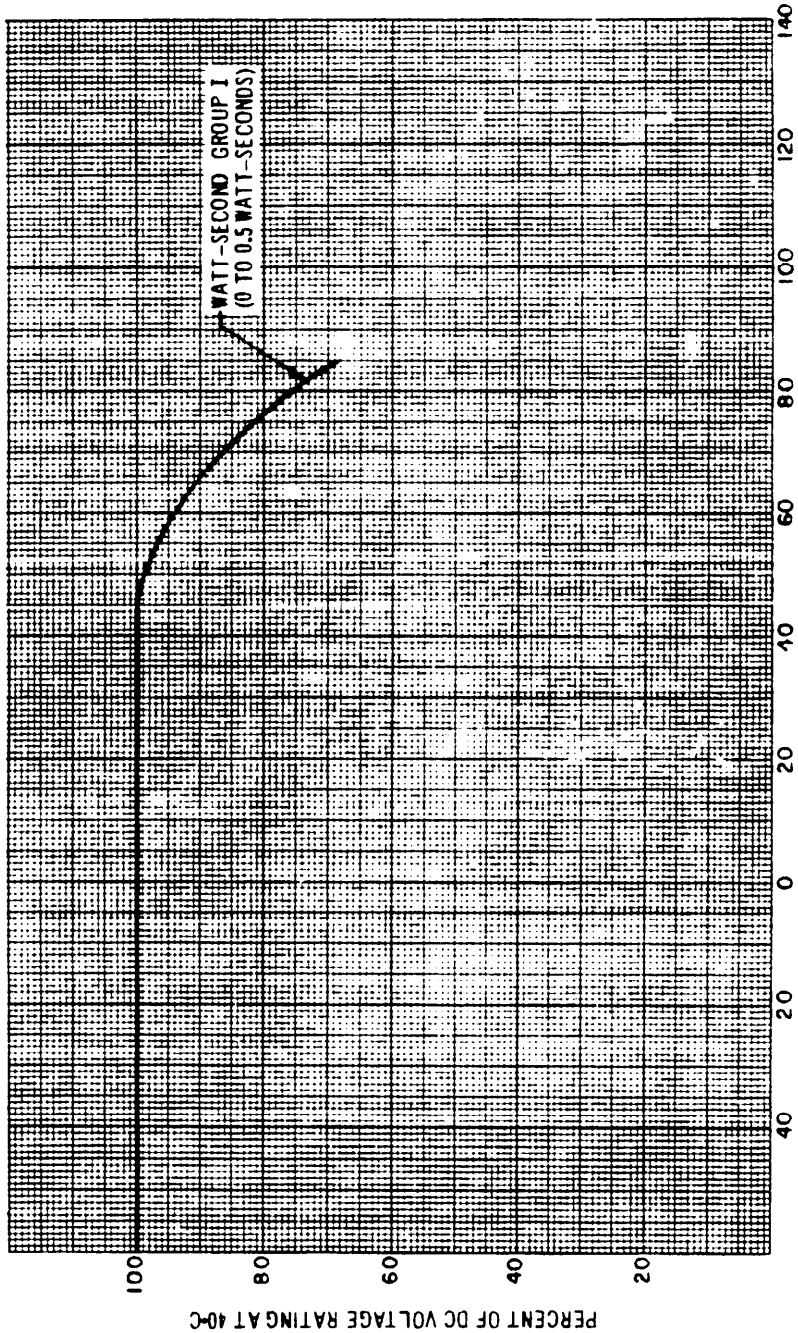
[4/] For quality conformance inspection, applications may be made between each terminal individually and the case, at the option of the supplier.

[5/] For capacitors having dc voltage ratings of 600 volts and less.

[6/] For capacitors having dc voltage ratings greater than 600 volts.

[7/] For quality conformance inspection, the application may be made for 1 second if the capacitor has passed group A inspection.

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NOTE. Permissible operating voltages, while based on incomplete data, are the capacitor suppliers' best estimate to provide a life expectancy of 8,800 hours of continuous operation at higher ambient temperatures. Longer life can be expected by operation at voltages lower than indicated on the curve; for example, a life expectancy of 44,000 hours may be obtained by operation at 70 percent of the voltage determined by use of this curve. Also, a life longer than 8,800 hours may be expected at the voltage determined by use of this curve if the high ambient temperature prevails for only a portion of the whole operating time.

FIGURE 2. Voltage derating for ambient temperature (characteristics E and F).

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TABLE V. Terminal to terminal insulation resistance measurements.

Capacitance values	Minimum insulation resistance [1]	
	Characteristic E	Characteristic F
0.33 microfarads and less - - - - - -	At 25° C 6,000 megohms	At 25° C 4,500 megohms
greater than 0.33 microfarads - - - -	2,000 megohm-microfarads [2]	1,500 megohm-microfarads [2]
greater than 0.033 microfarads - - -	At 85° C 20 megohm-microfarads [2]	At 85° C 15 megohm-microfarads [2]

[1] A potential equal to the dc rated voltage or 500 volts dc, whichever is less, shall be applied.

[2] Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Vibration (grade 1, 10 to 55 Hz, incl):
Method 201 of MIL-STD-202.
Duration - 3 hours (60 minutes in each direction).
Measurements - An electrical signal of 1±0.2 kHz, at a level of 1±0.5 volts applied during the last 30 minutes in each direction. There shall be no momentary arcing or breakdowns, open- or short-circuiting, or mechanical damage.

Salt spray (corrosion): Method 101 of MIL-STD-202, test condition B (48 hours), with no evidence of corrosion or obliteration of marking.

Temperature cycling: Method 102 of MIL-STD-202, test condition D.

Immersion: Method 104 of MIL-STD-202, test condition C, except that the duration of each immersion shall be 30 minutes, and change between solutions shall be made in not more than 3 seconds.
Measurements after final cycle:
Dielectric withstanding voltage, at 25° C:
Terminal to terminal - As specified in table IV.
Terminal to case (when case is not a terminal) - As specified in table IV.
Insulation resistance at 25° C:
Terminal to terminal - Not less than 30 percent of the value specified in table V.
Terminal to case (when the case is not a terminal) - Not less than 1,500 megohms.

Terminal strength: Method 211 of MIL-STD-202, test conditions A and E.
Test condition A (pull): Applied force - 5 pounds.
Test condition E (torque) - 1 pound-inch.

Moisture resistance: Method 106 of MIL-STD-202.
Polarization voltage - 100 volts.
Final measurements:
Dielectric withstanding voltage and insulation resistance requirements - As specified in immersion.

Seal: Method 112 of MIL-STD-202, test condition A, except that bath temperature shall be $+85^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C.
Liquid-filled capacitors - 10 minutes on each of two opposite sides at $+85^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C.
There shall be no evidence of leakage.

Low temperature and capacitance change with temperature:
Low temperature -
Conditions: -55° ± 3° C for 48 ± 4 hours with rated dc voltage applied.

Capacitance change with temperature:
Characteristic E - Not to exceed ± 10 percent.
Characteristic F - Not to exceed +10, -30 percent.

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Life: Method 108 of MIL-STD-202, test condition B (250 $^{+8}_{-0}$ hours).

Electrical load: 120 percent of rated dc voltage.

Final measurements (at 25° C):

Insulation resistance:

Terminal to terminal: Not less than 30 percent of the value specified in table V.

Capacitance: Within 10 percent of initial value.

Dissipation factor:

Characteristic E - Change not more than 1 percent from initial value.

Characteristic F - Change not more than 1.5 percent from initial value.

Inspection lot:

An inspection lot may consist of capacitors of the same characteristic described herein and the same characteristic of style CP91.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 11

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